

CREMS Mid Term Review Report

Car and Rental Equipment Management System

Carpenters Fiji Pte Limited

CS400 Industry Experience Project

Semester 2 2026

Group 10

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Executive Summary

CREMS has progressed from a project plan and requirements baseline to an integrated development prototype for Carpenters Motors and Carptrac. It connects customer enquiries, asset records, quotations, approvals and hire operations through a shared web application and database. Configurable divisions, branches and categories support later expansion. [2, 3]

The current build includes customer and staff interfaces, scoped access, booking workspaces, approval stages and guided pickup and return. Recent corrections address inspection photo attachment, inspection persistence, session renewal and customer catalogue images. Development verification provides evidence of these corrections, but does not constitute client acceptance or production readiness. [4]

The immediate priority is a reliable end-to-end vehicle and equipment hire journey. Remaining work includes maintenance and expenditure, equipment variations, reporting, client user acceptance testing and handover. The baseline completion date remains 30 October 2026. [2]

Document control and assistance

Version 1.1, updated from the 8 September draft for Group 10 academic and client review. Personal customer records, identity documents and signatures are excluded.

AI assistance was used to consolidate project documents, draft and format this report, and assist prototype development and debugging. The submitting team remains responsible for factual verification and assessment compliance. The assignment permits selective AI use for prototype development where students can explain and modify the work. [1]

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1 Project Context and Scope

Business problem

Carpenters Fiji manages rental vehicles and equipment with different operating rules. Vehicle hire involves licence eligibility, fuel and odometer readings. Equipment may require capacity checks, operators, transport, engine-hour tracking and additional approval. Separate operational records make it harder to establish availability, reconstruct condition at handover and relate asset income to maintenance costs. [2, 3]

CREMS provides a shared record across the hire lifecycle. A booking connects the customer, branch, asset and hire period. Inspections and agreements remain associated with that booking, while the asset register provides the longer-term asset history. A general asset inspection therefore does not establish the condition of a particular customer hire.

Objectives

The project aims to provide secure access, reliable availability and booking records, configurable pricing and approval, evidence-based pickup and return, and operational reporting. Success means an authorized user can complete a business task and retrieve its saved result within the correct organizational scope.

Included in the delivery scope	Outside the MVP or separately controlled
Motors vehicles and Carptrac equipment; configurable branches, categories and attributes.	Additional divisions require operational validation and their own acceptance work.
Customers, assets, availability, quotes, bookings, approvals and hire documents.	Full accounting/ERP replacement, payroll and banking reconciliation.
Inspection evidence, handover, return, maintenance and expenditure.	Live telematics, advanced route optimization and native offline applications.
Operational reports, audit, notifications and profitability measures.	Production payment gateway, third-party identity and large historical migration unless approved.

Stakeholders and expected value

Customers need clear charges and booking status. Rental officers need reliable queues; managers need scoped oversight and approval evidence. Maintenance staff need service and condition history. Client sponsor Amit Kumar and academic supervisor Dr Ravneil Nand provide business and academic review. [2]

Expected benefits include fewer duplicated records, clearer handover responsibility and better visibility of asset use and cost. These remain objectives: measured reductions in processing time or operating cost have not yet been established.

2 Functional Requirements

The requirements below connect the functional baseline to acceptance scenarios. Development status describes implementation rather than client sign-off. Acceptance requires the expected business outcome, persistence and authorization to be verified. [2–4]

ID and requirement	Current position	Required acceptance evidence
FR01 Identity and access	Implemented; wider security verification remains.	Customer sees own records only; staff role and branch restrictions hold at API level.
FR02 Customer records	Portal and staff administration demonstrable.	Create and update customer; retrieve permitted documents and reject unauthorized access.
FR03 Assets and catalogue	Implemented; image recovery verified locally.	Persist attributes, branch and status; display catalogue photos correctly.
FR04 Availability and requests	Booking and quote flows implemented and refining.	Reject unavailable hire; retain dates, branch and request reference.
FR05 Pricing and quotations	Pricing workspace and quote decisions implemented.	Charges, tax, deposit and revisions match agreed examples.
FR06 Configurable approvals	Ordered stages and scoped decisions implemented.	Triggered hire completes required stages before confirmation.
FR07 Pre-hire and checkout	Guided flow and independent saving implemented; targeted tests pass.	Save and reopen photos/checklist without checkout; complete handover separately.
FR08 Return and post-hire	Return and comparison implemented; browser payload checks completed.	Retain evidence; apply correct asset status and damage referral.
FR09 Maintenance and costs	Foundation exists; operational completion remains.	Record job costs and downtime; verify return-to-service control.
FR10 Reports and notifications	Partial; reconciliation and delivery verification remain.	Reconcile reports to transactions; verify delivery outcome and audit history.

Traceability and change control

Each scenario should retain its requirement ID, test data, expected and actual results, defect reference and reviewer. A changed client rule must update both requirement and test. The matrix supports verification without treating a visible screen as proof of correctness.

3 Quality and Ethical Requirements

Quality targets

The plan sets two-second targets for routine interactions and five-second targets for longer operations, with at least 80 percent unaided completion of core tasks in moderated usability testing. These are acceptance targets; no project-wide performance or usability result is asserted here. [2]

Area	Required control and verification
Access and sessions	Enforce role and branch scope on the server. Test expiry, active renewal, logout and prohibited access using separate accounts.
Integrity	Validate dates, meters, payments and lifecycle transitions. Verify persistence after reload and conflicting operations.
Files and privacy	Restrict photo types and sizes; protect private documents and signatures. Verify download authorization and agree retention rules.
Recovery	Maintain migrations and recoverable configuration. Rehearse backup restoration and clean setup before handover.
Usability	Use labelled controls, clear errors and consistent navigation. Verify keyboard access, responsive layouts and representative tasks.

Handling personal information and condition evidence

Identity documents, photos and signatures can expose personal information. Demonstrations should use non-sensitive data and staff access should match the role’s purpose. Inspection photographs should show the asset without unnecessary faces, documents or unrelated property. Storage duration and permitted recipients need client agreement; upload limits alone do not resolve these responsibilities.

Condition evidence can affect damage disputes and charges. Before-and-after records must retain their booking association and responsible staff identity. Users need to distinguish a saved preparation record from completed handover. A damage allegation should remain subject to review rather than being treated as proven by a checkbox or note.

Transparency and professional responsibility

Estimates should distinguish rental charges, extras and refundable bonds. Approval rules must be applied consistently and changes must be auditable. The team must accurately describe the prototype’s limits. Client material should use approved distribution channels, and external sources must be attributed. No legal compliance certification is claimed. [1, 2]

4 System Architecture and Data Model

The web application uses React 19, TypeScript, Vite and Material UI. An ASP.NET Core API on .NET 10 applies business rules, and Entity Framework Core persists records in SQL Server. The development database runs in Docker. [3, 4]

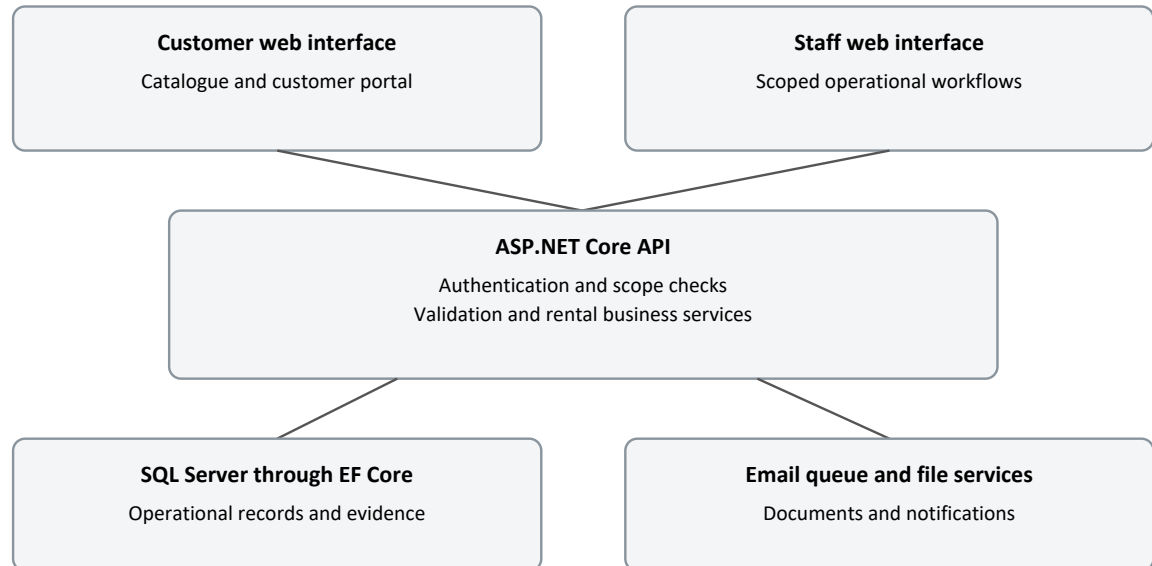


Figure 1 Logical architecture of the development system

The API validates the user, scope and business state before protected changes. Browser validation improves usability but cannot replace server controls. Supporting file and notification services operate alongside the transaction.

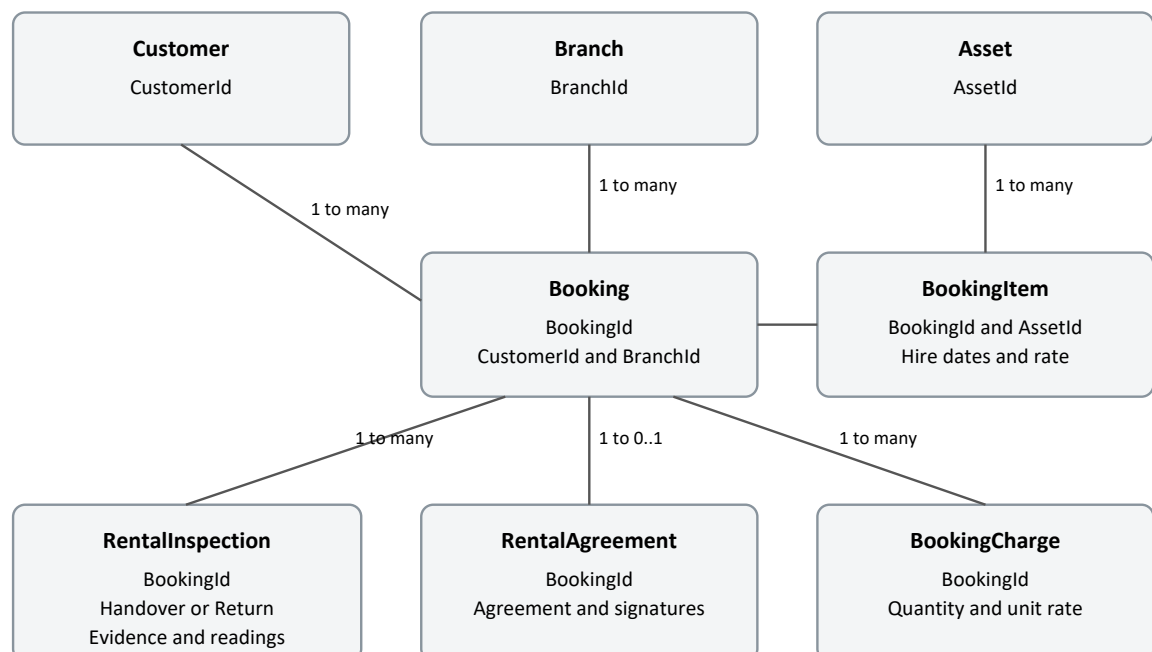


Figure 2 Selected rental entities and relationships

BookingItem links each asset and hire period to a booking. RentalInspection stores handover or return evidence. One booking can have many items. Supporting approval, identity, payment and asset-attribute entities are omitted for readability. This is a simplified technical model, not an approved full ERD. [4]

5 User Responsibilities and Workflow Design

Actor	Principal use cases
Visitor and Customer	Browse rentals; request booking or quotation; manage own profile and monitor own requests and documents.
Rental Officer	Review requests, prepare pricing, allocate assets and perform authorized pickup, inspection and return.
Branch Manager	Review branch activity and decide assigned approval stages within permitted scope.
Maintenance Officer	Review asset condition and carry out authorized maintenance and service updates.
Administrator and Super Administrator	Manage permitted accounts and configuration; reserve system-wide governance for appropriate authority.

The revised role baseline removes separate Driver or Operator and Finance Officer login actors. Personnel requirements remain rental resources and charges without requiring unnecessary account access. [3]

Pickup and inspection sequence

A confirmed booking opens the existing Pickups & returns workflow from Bookings & quotations. Staff confirm the customer, scan the asset, verify documents, record meter and fuel, complete the pre-hire inspection, review the agreement, obtain the customer signature and perform agent checkout. Authorized operational staff conduct the inspection.

The pre-hire step saves a preparation record independently. Reopening should restore checklist, photos and readings. Saving must not change the booking into an active hire; checkout is a separate action with its own readiness checks. General Asset register inspections remain asset records. Rental pre-hire and post-hire evidence belongs to the booking workflow. [4]

Return and equipment variations

At return, staff record condition and readings, compare pre-hire evidence and record damage and accessories. The next asset state depends on the result and maintenance needs. Vehicle-specific licence and meter fields must not block equipment where they do not apply. Equipment acceptance also requires appropriate operator, engine-hour and safety scenarios.

Project-specific design contribution

Shared workflows use configurable categories, checklist templates and staged approval to support different asset classes while preserving consistent evidence ownership. Reusing the operational page for booking and QR entry points reduces duplicate forms and inconsistent validation.

6 Milestones and Engineering Evidence

The twelve-week baseline runs from 10 August to 30 October. At 16 September, delivery is in Week 6. Course assessment weeks and project sprint weeks are separate schedules. Some lifecycle functionality was implemented early; its milestone still depends on integrated demonstration. [1, 2]

Baseline milestone	Position at review
M1 Week 1 Scope and foundation	Plan, repository, API, web application and database available. Approval records remain part of project governance.
M2 Week 2 Secure sign-in	Authentication and sessions implemented; active-session renewal corrected during testing.
M3 Week 3 Access isolation	Role and branch scope implemented; full isolation regression remains required.
M4 Week 5 Asset register	Asset/category interfaces and catalogue available; targeted image checks completed.
M5 Week 7 Quotation to booking	Booking and staged approvals implemented; complete notification and agreement demonstration remains.
M6 Week 8 Motors lifecycle	Guided pickup/return implemented early; targeted defects corrected and tests added.
M7 Week 10 Carptrac lifecycle	Equipment foundation available; personnel and engine-hour acceptance remains.
M8 Week 12 Handover	Deployment, final acceptance, guides and closure remain planned.

Completed development verification

The latest backend run passed 31 tests. A database check inserted an inspection with photo and checklist evidence, cleared the tracked context, read it back, updated it and confirmed unchanged booking status. The transaction was rolled back, leaving no test inspection. This verifies persistence without completing a customer hire. [4]

Browser component checks covered selection, resizing, corrupt-file errors, previews, pickup/return payloads and preparation save/reopen behaviour. API writes were mocked for these checks. Separately, all 32 restored catalogue image URLs responded successfully through the frontend and 12 visible photos decoded in headless Chrome. These results do not replace authenticated end-to-end tests or client UAT. [4]

7 Plan Changes and Issues Resolved

Changes from the baseline

Configurable approval and parts of the rental lifecycle were developed ahead of the baseline sequence, exposing integration problems early. Remaining time will be used for stabilization and complete business scenarios while protecting testing and handover. The draft also narrows initial customer handling and removes unnecessary login roles. These changes require continued client review. [2, 3]

Observed issue	Engineering response	Verification and remaining work
Inspection could not save	Register new inspections explicitly for insertion; reuse the preparation record on subsequent saves.	Regression tests and rolled-back SQL insert/read/update passed; user workflow retest remains.
Photo attachment unreliable	Shared picker with decoding, resizing, limits, previews and processing status.	Browser checks cover valid and corrupt files; real device and camera checks remain.
Save could lead to login	Renew server session during active use while retaining inactivity expiry.	Controlled-clock tests cover active/idle behaviour; full session regression remains.
Catalogue images broken	Restore missing bundled seed images and avoid stale failed-image cache responses.	32 URLs passed; 12 visible photos decoded.
Booking inspection entry confusing	Redirect to existing Pickups & returns with the selected booking.	Focused booking and preparation checks added; no separate inspection page needed.
Equipment pickup had vehicle assumptions	Allow non-applicable driver/meter fields while retaining applicable readiness checks.	Targeted request and workflow tests; full equipment acceptance remains.

Lessons and unresolved work

A success path is insufficient unless the record survives reload. Database tracking, cookie expiry and file decoding can each break an apparently complete form. Verification now includes stored evidence and failure handling. Broader integration checks and branch-scoped administration errors still need triage before release; the corrections do not establish that every module is defect-free.

8 Risk Management Update

The plan scores probability and impact from 1 to 5. Existing scores are retained below: a completed mitigation does not justify a lower score without review. Observed defects are issues; recurrence and delivery consequences remain risks. [2]

Risk and baseline score	Mitigation and current position	Accountable role
Scope/capacity R1/R3 20 High	Maintain initial division boundary; defer optional scope before reducing verification time.	Project Leader
Client availability R2 16 High	Book focused reviews early; log assumptions and request explicit operational decisions.	Project Leader
Scope leakage R4 15 High	Scoped queries and API policies exist; cross-branch negative tests remain necessary.	Backend lead
Invalid booking state R5 15 High	Validate transitions and persistence; test concurrency and duplicate submission.	Backend lead
Pricing R6 15 High	Reconcile decimal calculations and line items against client-approved examples.	Backend lead
Photo privacy R7 12 Medium	Type/size controls implemented; review private access, retention and storage growth.	Quality lead
Late mismatch R12 15 High	Demonstrate complete journeys; block handover on unresolved critical defects.	Project team
Test backlog R21 12 Medium	Targeted tests added; extend integration coverage before optional features.	Quality lead

Additional risks identified

Incorrect handling of new related entities could recur in other save operations. Session expiry during long forms threatens work continuity. Missing media files can make otherwise valid asset records unusable. These scenarios call for persistence regression, realistic form-duration tests and deployment media checks. Backend and quality leads are proposed owners; probability and impact should be recorded at the next team review.

No risk closure is claimed solely from a local correction. Recovery rehearsal, production configuration review, real-device capture and client acceptance remain necessary before release.

9 Remaining Delivery Plan

The working sequence retains the baseline end date and maintenance, equipment and handover milestones. Early implementation reduces integration uncertainty. This schedule is proposed for team and client review; it does not represent new client sign-off. [2]

Period	Deliverable and lead	Exit evidence
16–18 Sep	Stabilize rates, quotes and inspection saving. Backend and Quality leads.	Record survives reload; pricing matches examples; defects triaged.
21–25 Sep	Quotation-to-booking, approvals, agreements and notifications. Project and Backend leads.	Standard and triggered requests complete with correct references, documents and delivery outcome.
28 Sep–2 Oct	Motors lifecycle validation. Frontend and Quality leads.	Booking completes pre-hire, QR handover, return and post-hire with correct evidence and state.
5–9 Oct	Maintenance, damage and expenditure. Backend lead.	Job costs, downtime, referral and return-to-service restrictions verified.
12–16 Oct	Carptrac variations and UAT preparation. Backend and Frontend leads.	Operator, engine-hour, transport and safety cases pass; UAT scripts ready.
19–23 Oct	Reports and integrated UAT. Quality lead and client reviewers.	Reports reconcile; critical journeys pass; feedback corrected or formally deferred.
26–30 Oct	Stabilization and handover. Project Leader and all members.	Blocking defects resolved; restore/setup proven; guides, release and final assessment materials delivered.

Critical dependencies

Reports depend on trustworthy rental and expenditure records. Return-to-service depends on inspection and maintenance outcomes. The team will prioritize these transactions before accepting derived dashboards. A critical-path delay requires escalation with an explicit scope or schedule decision.

Release criteria

Critical business-rule and API tests must pass, high-severity security issues must receive approved treatment, and agreed browser journeys must succeed. Client feedback must be resolved or explicitly accepted. Performance and usability targets need measurement, and deployment and restoration procedures need demonstration. [2]

10 Teamwork and Project Control

Responsibility baseline

The plan assigns four primary roles, with shared review, testing and presentation responsibilities. These are ownership assignments, not claims of individual hours or completed contributions. [2]

Member	Primary responsibility
Kavish Chandra	Project Leader/client liaison: scope, decisions, coordination, integration and submission.
Shoneel Kumar	Backend/data lead: API and domain model, database, migrations and authorization.
Sudhansu Jayshil Kisun	Frontend/UX lead: journeys, interface integration, accessibility and UI verification.
Rahul Chand	Quality/DevOps lead: tests, builds, environments, release, backup and documentation.

Communication and participation

Weekly reviews connect client feedback to requirements, defects and the next increment. Tasks should identify an owner and retain code or document evidence. Shared demonstrations allow members to explain related modules and give the client an opportunity to correct assumptions early.

Requirements, diagrams, workflow explanations and prototype evidence support communication to technical and non-technical audiences. Team participation should be substantiated through meeting notes, reviewed changes and contribution records, rather than inferred from the report alone. [1, 2]

Resources and cost control

The baseline assumes four members at 6.5 hours each per week for twelve weeks, or 312 planned team hours. The maximum planning budget is FJD 580. These are allowances, not actual time or expenditure records. Free-tier resources remain the default, with spending subject to the plan's approval process. [2]

Change and knowledge control

Material changes need a recorded business reason, schedule impact and decision before inclusion. Git history, reviewed changes and setup guidance support recoverability. Security, persistence and pricing decisions should be explained across the team so knowledge is not concentrated in one member.

Before presentation, each member should be able to explain the requirements, demonstrate the assigned workflow and accommodate a requested prototype change. Final report review and understanding remain the submitting team's responsibility. [1]

11 Conclusion and References

CREMS has an integrated development foundation and operational interfaces for the initial Motors and Carptrac scope. Recent corrections address faults that prevented inspection evidence from being attached, saved or displayed reliably. Targeted tests support those corrections, including database persistence, while full client acceptance remains ahead.

The next phase should establish verified customer-to-return journeys for both divisions, then complete maintenance and reporting against reliable transaction records. The 30 October target remains a planning objective dependent on scope control and protected verification and handover time.

References

- [1] The University of the South Pacific. *CS400 Industry Experience Project Semester 2 2026 Assignment 2 Mid-Term Review*. Supplied assessment brief, 6 pages. Requirements and submission format, pp. 1–2; report and learning-outcome rubrics, pp. 3–4.
- [2] CREMS Project Team. *Project Management Plan*. Semester 2, 2026. Supplied project document. Scope, milestones and schedule, printed pp. 3–5; quality, pp. 18–20; risks, pp. 21–25; resources, p. 26.
- [3] CREMS Project Team. *CREMS Mid Term Review Report*. Version 1.0, 8 September 2026. Supplied draft, CREMS_Mid_Term_Review_Report_2026_1.docx. Business context, revised actor model and reported prototype progress.
- [4] CREMS development repository and local verification records, through 16 September 2026. Implementation: Booking.cs, RentalInspectionPersistence.cs, InspectionPhotos.tsx, useRentalInspection.ts and guided rental workflows. Verification: InspectionTrackingTests.cs, browser workflow checks and the rolled-back inspection database check.

Evidence interpretation

The plan and draft support project context and reported progress. Local development records support the specific engineering observations. They do not establish client acceptance, production deployment, measured business savings or comprehensive security certification. External works listed in the plan are not reproduced as if directly consulted for this review.

Next client review

Confirm inspection checklists, equipment readiness, damage handling and semester reporting measures. Record acceptance results for an ordinary vehicle hire and an approval-triggered equipment hire, including saving and reopening inspection evidence before handover.